

Relations II

Previous Lecture

- *Cartesian product*
- *Properties of Cartesian products*
- *Relations*

Describing Relations

- *The simplest way to describe or represent a relation is by its set of ordered tuples*
- *Example:*

Among 6 people, Mark, Jerry, John, Randy, Aaron, and Ralph, Mark and Randy are brothers, and also John, Aaron and Ralph are brothers

$$A = \{ \text{Mark, Jerry, John, Randy, Aaron, Ralph} \}$$
$$B = \{ (x,y) \mid x \text{ is a brother of } y \}$$
$$= \{ (\text{Mark,Randy}), (\text{Randy,Mark}), (\text{John,Aaron}), (\text{Aaron,John}), (\text{John,Ralph}), (\text{Ralph,John}), (\text{Aaron,Ralph}), (\text{Ralph,Aaron}) \}$$

Describing Binary Relations

- Can also describe *binary* relations by a *table* or *matrix*

Matrix of a relation $R \subseteq A \times B$ is a rectangular table, rows of which are labeled with elements of A (in any but fixed order), and columns are labeled with elements of B . We write 1 in the intersection of row a and column b if and only if $(a,b) \in R$; otherwise we write 0.

Brotherhood

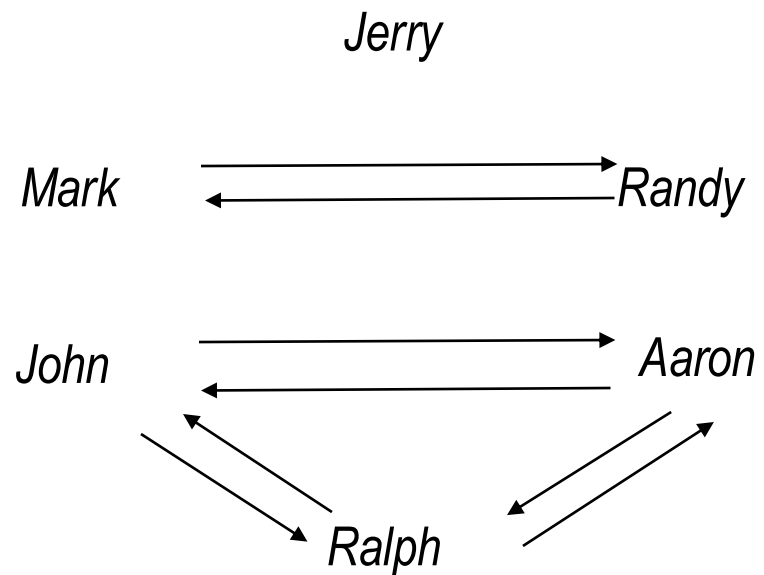
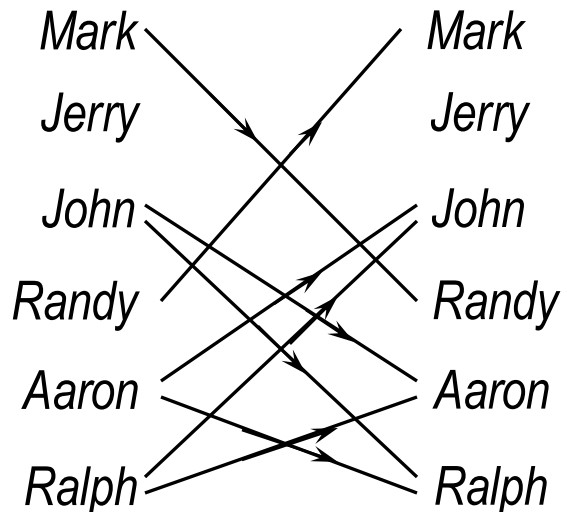
	Mark	Jerry	John	Randy	Aaron	Ralph
Mark	0	0	0	1	0	0
Jerry	0	0	0	0	0	0
John	0	0	0	0	1	1
Randy	1	0	0	0	0	0
Aaron	0	0	1	0	0	1
Ralph	0	0	1	0	1	0

Describing Binary Relations (cntd)

- Binary relations can also be described by a **graph**

Graph of a relation $R \subseteq A \times B$ consists of two sets of vertices labeled by elements of A and B . A vertex a is connected to a vertex b with an edge (arc) if and only if $(a,b) \in R$.

If $A = B$ then we may use only one set of vertices



Example

● Describe the relation $R(a,b) = \{ (a,b) \mid a,b \in \{1,2,3\} \text{ and } a \leq b \}$

Properties of Binary Relations – Reflexivity

- A binary relation $R \subseteq A \times A$ is said to be **reflexive** if $(a, a) \in R$ for all $a \in A$.

Example:

$(a, b) \in R \subseteq \mathbb{Z} \times \mathbb{Z}$ if and only if $a \leq b$

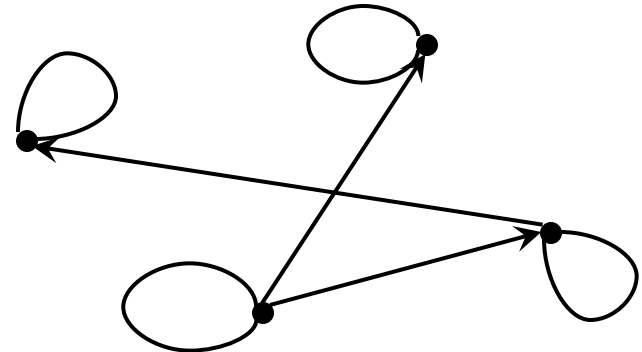
This relation is reflexive, because $a \leq a$ for all $a \in \mathbb{Z}$

Matrix:

$$\begin{pmatrix} 1 & * & * & * \\ * & 1 & * & * \\ * & * & 1 & * \\ * & * & * & 1 \end{pmatrix}$$

1's on the diagonal

Graph:



Loops at every vertex

Properties of Binary Relations – Transitivity

- A binary relation $R \subseteq A \times A$ is said to be **transitive** if, for any $a, b, c \in A$, if $(a, b) \in R$ and $(b, c) \in R$ then $(a, c) \in R$.

Example:

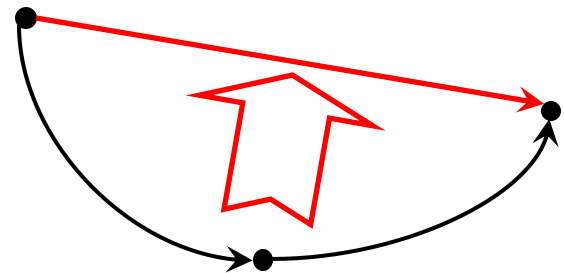
$(a, b) \in R \subseteq \mathbb{Z} \times \mathbb{Z}$ if and only if $a \leq b$

This relation is again transitive, because if $a \leq b$ and $b \leq c$, then $a \leq c$ for all $a, b, c \in \mathbb{Z}$

Matrix:

?

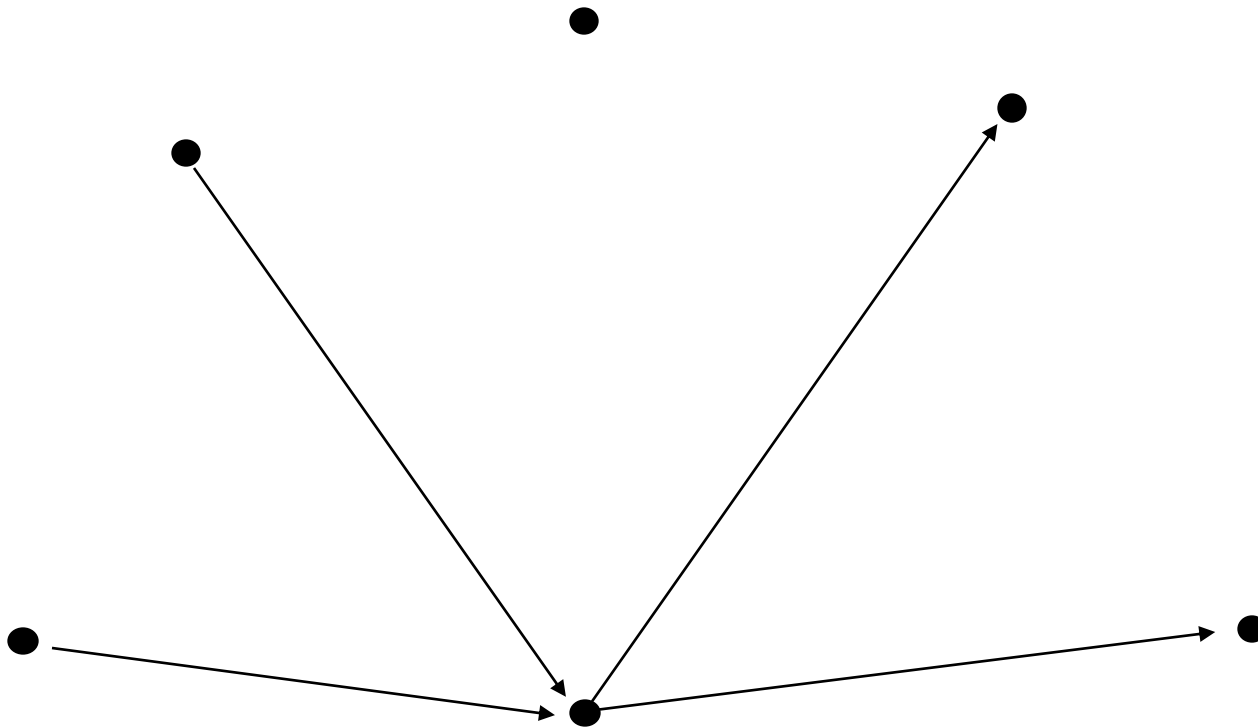
Graph:



Edges compose

Exercise

● *Make the following graph reflexive and transitive*



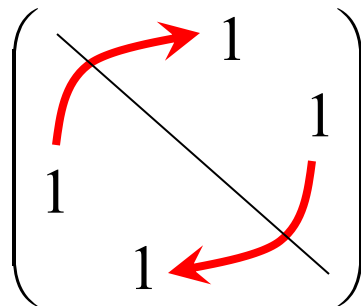
Properties of Binary Relations – Symmetricity

- A binary relation $R \subseteq A \times A$ is said to be **symmetric** if, for any $a, b \in A$, if $(a, b) \in R$ then $(b, a) \in R$.

Example:

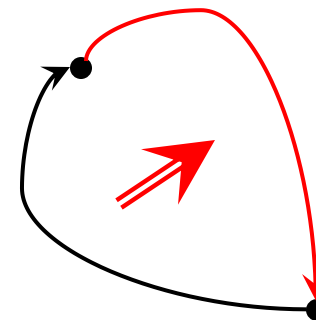
The relation $a = b$ is symmetric because if $a = b$, clearly also $b = a$

Matrix:



Matrix is symmetric w.r.t. the diagonal

Graph:



Every edge has an opposite

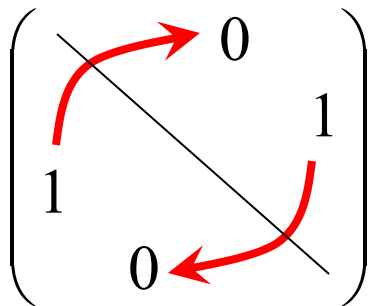
Properties of Binary Relations – Anti-Symmetry

- A binary relation $R \subseteq A \times A$ is said to be **anti-symmetric** if, for any $a, b \in A$, if $(a, b) \in R$ and $(b, a) \in R$ then $a = b$.

Example:

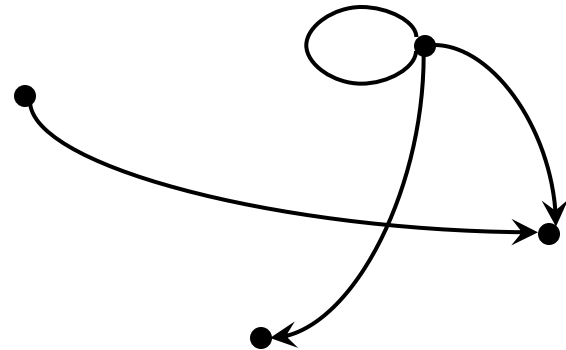
The relation $a < b$ is anti-symmetric because if $a < b$, b cannot also be less than a

Matrix:



*Matrix is anti-symmetric w.r.t.
the diagonal*

Graph:



No edges have an opposite

Examples

	<i>reflexive</i>	<i>symmetric</i>	<i>transitive</i>	<i>anti-symmetric</i>
<i>Brotherhood</i> <i>x is a brother of y</i>	<i>It depends</i>	Yes (on men) No (on people)	No (with half brothers) Yes (otherwise)	No
<i>Neighborhood</i> <i>x is a neighbor of y</i>	<i>It depends</i>	Yes	No	No
$x \leq y$	Yes	No	Yes	Yes
$x = y$	Yes	Yes	Yes	No

Equivalence relations

- A binary relation R on a set A is said to be an **equivalence relation** if it is **reflexive**, **symmetric**, and **transitive**.

- Example:

Let $R = \{(a,b) \subseteq \text{People} \times \text{People} \mid a \text{ \& } b \text{ are the same age.}\}$

Reflexive?

Symmetric?

Transitive?

- Example:

Let $R = \{(a,b) \mid a \leq b.\}$

Reflexive?

Symmetric?

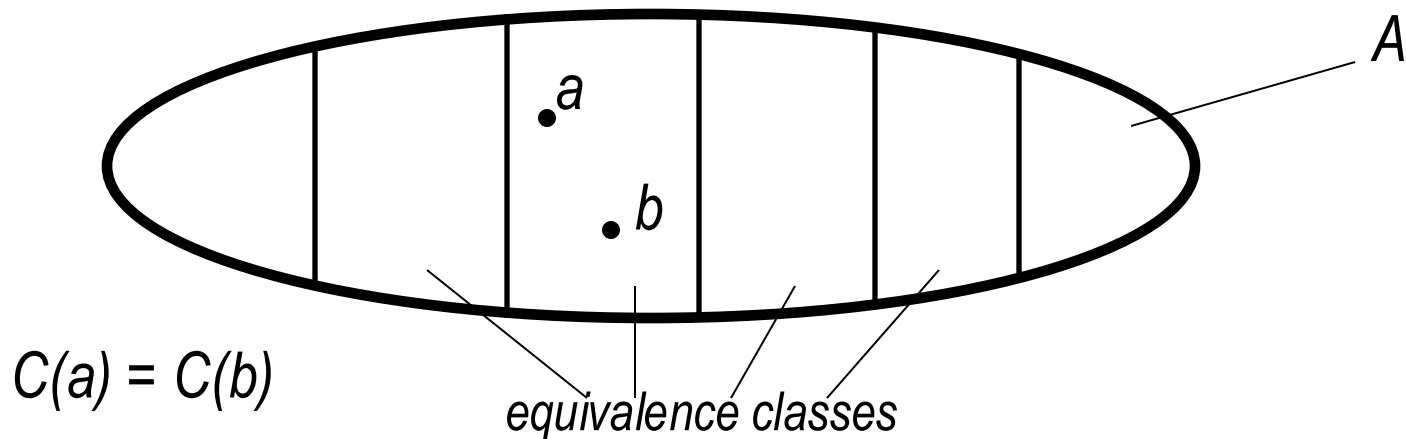
Transitive?

Equivalence Classes

- Take $a \in A$ and R an equivalence relation on A .
- The set $C(a) = \{ b \mid (a,b) \in R \}$ is the **equivalence class** of a .
- Informally, $C(a)$ is "all elements which are equal to a wrt R "
- Example:
 - Let $R = \{(a,b) \subseteq \text{People} \times \text{People} \mid a \text{ \& } b \text{ are the same age.}\}$
 - $C(\text{Isaac}) = \{ b \mid b \text{ is 3 years old} \}$
 - $C(\text{Elliot}) = \{ b \mid b \text{ is 1 year old} \}$

Partitions

- Thus the equivalence classes divide up the set A into disjoint subsets.



- A collection of subsets $M_1, \dots, M_n$ of a set A is called a **partition** if the following conditions hold.

- (1) Every $M_i \neq \emptyset$
- (2) If $M_i \neq M_j$ then $M_i \cap M_j = \emptyset$
- (3) $A = \bigcup_{i=1}^n M_i$



Partitions and Equivalence Relations

● **Theorem.** *Let A be a set.*

(1) If R is an equivalence relation on A , then its equivalence classes form a partition of A .

(2) If $M_1, \dots, M_n$ is a partition of the set A , then the relation R defined as follows: $(a, b) \in R$ if and only if $a, b \in M_i$ for some M_i , is an equivalence relation on A .

● *Proof*

(1) Next slide

(2) Homework

Equivalence classes partition a set

● Let R be an equivalence relation on A . We need to show

- (1) For any $a \in A$, the class $C(a) \neq \emptyset$
- (2) If $C(a) \neq C(b)$ then $C(a) \cap C(b) = \emptyset$
- (3) $A = \bigcup_{a \in A} C(a)$

● **Proof**

- (1) R is reflexive, therefore, $(a, a) \in R$. Hence $a \in C(a) \neq \emptyset$
- (3) Clearly $\bigcup_{a \in A} C(a) \subseteq A$ because each $C(a) \subseteq A$
Now suppose $a \in A$. Then $a \in C(a)$, so $a \in \bigcup_{a \in A} C(a)$
and hence also $A \subseteq \bigcup_{a \in A} C(a)$

Equivalence classes partition a set (cntd)

Proof of (2): *If $C(a) \neq C(b)$ then $C(a) \cap C(b) = \emptyset$*

Suppose $c \in C(a) \cap C(b)$ to prove by contrapositive

We need to show that $C(a) = C(b)$.

First we show that $(a,b) \in R$:

Since $c \in C(a) \cap C(b)$, we have $(a,c), (b,c) \in R$.

Since R is symmetric, $(c,b) \in R$.

Since R is transitive, $(a,b) \in R$.

Now we show $C(a) \subseteq C(b)$

Let $x \in C(a)$. Then $(x,a) \in R$

But $(a,b) \in R$, so by transitivity, $(x,b) \in R$

Hence $x \in C(b)$

Remaining case $C(b) \subseteq C(a)$ is similar

Q.E.D.

Example: congruences (more in a few weeks)

- Let k be an integer. Integers a, b are **congruent modulo k** , denoted $a \equiv b \pmod{k}$, if their remainders when they are divided by k are equal, or, equivalently, if k divides $a - b$.
 ... -3, 0, 3, 6, ... are congruent modulo 3,
 and so are ..., -4, -1, 2, 5, ... and ..., -5, -2, 1, 4, ...
- The relation $\equiv \pmod{k}$, 'to be congruent modulo k ' is
 - reflexive, because k divides $a - a = 0$
 - symmetric, because if k divides $a - b$ then it also divides $b - a$
 - transitive, because if k divides $a - b$ and $b - c$, then it also divides $a - c = (a - b) + (b - c)$
- $\equiv \pmod{k}$, is an equivalence relation with equivalence classes $\{a \mid \text{there is } b \text{ with } a = bk + c\}$
- Arithmetic on these classes is called **modular arithmetic**